

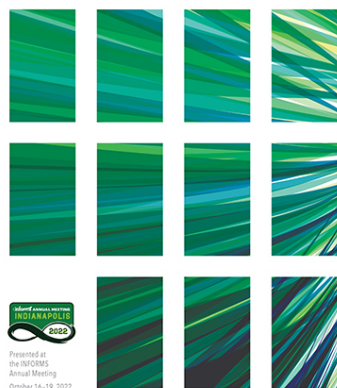


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Choosing Opponents in Tournaments

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Choosing Opponents in Tournaments

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Abstract We consider the design of tournaments that use a preliminary stage, followed by several rounds of single elimination play. Many sports, including most U.S. major sports, use this format. The tournament design problem involves determining the sequence of matchups that occur in order to determine a winner. This problem has been extensively studied within the operations research and economics literature. The conventional design of the single elimination rounds is a “bracket” based on the players’ prior ranking or seeding. However, this design suffers from several deficiencies, insofar as natural expectations about the results of the design are not satisfied. First, the expectation that higher-ranked players have a higher probability of winning is not satisfied. Second, the probability that the top two players meet is not maximized. Third, there is the widely observed issue of tanking or shirking at the preliminary stage, where there are disincentives for a player to win. Fourth, top-ranked players randomly incur unfortunate matchups against other players, which introduces an unnecessary element of luck. Finally, the use of a conventional fixed bracket is limiting in that it fails to allow players to consider information that develops during the tournament. Our proposed solution allows higher-ranked players at the single elimination stage to choose their next opponent at each round. Using data from 1,902 men’s professional tennis tournaments during 2001–2016, we demonstrate the reasonableness of the results obtained. We also perform sensitivity analysis for the effect of increasing irregularity in the pairwise win probability matrix on three traditional performance measures. Finally, we consider strategic shirking behavior and show how our opponent choice design can eliminate or reduce such behavior, including some famous examples. In summary, compared with the conventional design, the opponent choice design provides higher probabilities that the best player wins and also that the two best players meet, and it reduces strategic shirking by both individual players and groups.

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Keywords sports tournament • opponent choice • performance criteria • ranking • strategic behavior • shirking • professional tennis data • computational study • sensitivity analysis

1. Introduction

We consider tournaments that consist of a preliminary or group stage, followed by several rounds of single elimination play. Such tournaments are used extensively to organize sports events worldwide. Our work applies to both individual and team tournaments; hence we use the term “player” to describe either an individual player or a team. A ranking or seeding is established prior to the first round of single elimination play. This seeding determines the

sequence of opponents each player must face in order to progress through the tournament. The information that determines this seeding can come from the preliminary or group stage or be based on previous performance, such as over an entire season of play. Tournaments that use this type of design include the championships of most U.S. major sports, the FIFA (Fédération Internationale de Football Association) World Cup, and the International Cricket Council's World Cricket Cup.

The designers of such tournaments have several objectives, typically based on financial motivation and public appeal. Objectives include providing the players with equally fair opportunities, motivating them to perform well, selecting among them, and providing an appealing event for spectators. From a longer-term perspective, the outcome of the tournament contributes to the appeal of future events and the career success of the players. Our work addresses several deficiencies that arise in such tournaments with respect to these objectives, as we now discuss.

1. As documented in the tournament design literature we discuss in what follows, the reasonable expectation that higher-ranked players have a higher probability of winning the tournament is not satisfied.
2. The probability that the top two players meet, which is a desirable outcome from the viewpoint of attracting public interest, is not maximized.
3. There is the widely observed issue of *shirking* or *tanking* at the preliminary stage, where a player deliberately avoids winning a game. This occurs when a preliminary stage match can be lost, but the player continues to the next round anyway and has an incentive to shirk in order to avoid more difficult opponents later in the tournament.
4. Top-ranked players may randomly incur matchups against other players that are unfortunate, in the sense that their performance records against such players are weaker than the general performance of both players would suggest. This occurrence introduces an unnecessary element of luck into the tournament. It seems reasonable to believe that the achievement of a top ranking at the preliminary stage should not disadvantage a player based on chance.
5. The use of a conventional tournament design fails to allow players to consider information that develops during the tournament, such as injuries to other players. As a result, this information cannot be used to adjust a player's strategy.

To address the deficiencies of the conventional tournament design enumerated above, we propose a new design under which the players, in a ranked order that is at least initially determined at the preliminary stage, *choose their next opponent at each single elimination round*. The choice of opponent is made under two conditions:

- i. The opponent has not been previously chosen by a higher-ranked player.
- ii. The player making the choice has not been previously chosen.

This opponent choice design allows considerable flexibility in implementation. For example, we study the performance of two versions: under *static ranking*, where the ranking of the players when they enter the single elimination stage of the tournament remains fixed, and under *dynamic ranking*, where a lower-ranked player inherits the ranking of a higher-ranked player, which the former then beats. Flexibility is also available regarding the number of player(s) from the top of the ranking that are allowed to choose their opponent(s). Another type of flexibility allows exemption of higher-ranked players from being chosen. Under every implementation we propose, the opponent choice design provides an easier path through the tournament for any player as a result of obtaining a higher ranking. In doing so, it largely eliminates the incentive for shirking.

This work makes the following contributions. We propose two versions of our opponent choice design—that is, static and dynamic—that address the above-mentioned deficiencies in conventional tournament design. We provide an algorithm that computes, for each player, a sequence of opponents to choose that maximizes its tournament win probability. We study

the performance of the opponent choice design using three established measures from the related literature. These measures are

- i. whether the opponent choice design provides a reasonably higher tournament win probability to the top-ranked player(s),
- ii. whether it provides a reasonably higher probability of the top two ranked players meeting, and
- iii. whether it provides a ranking by tournament win probabilities that closely matches the original ranking of the players.

As shown by our computational study in Section 5, the opponent choice design outperforms the conventional tournament design for the first two performance measures. For the third measure, the results are very similar to those from the conventional bracket design. We also study the robustness of our results by considering tournaments with incomplete information. Finally, we show that our opponent choice design eliminates or reduces incentives for shirking under both individual and group strategic behavior, and it provides several other advantages, including the promotion of public interest in the tournament.

The rest of this tutorial is organized as follows. Section 2 lists some current applications of opponent choice in tournaments and reviews the related literature. Section 3 defines the problem of optimal opponent choice to maximize a player's tournament win probability. Section 4 describes a general tournament round and provides an algorithm by which any player can maximize its tournament win probability, along with a set of real data and a numerical example. In Section 5, we first provide a computational study of tournament win probabilities using the tennis data. We also study the sensitivity of the three performance measures previously discussed to increasing irregularity in the players' pairwise win probability matrix. Section 6 discusses the performance of the opponent choice tournament design under two cases with incomplete information. In Section 7, we show that using opponent choice mitigates shirking. Section 8 provides a summary and suggestions for future work.

2. Applications and Literature

Section 2.1 describes several tournaments where opponent choice is currently used in practice. Section 2.2 reviews the literature of tournament design.

2.1. Applications

Several tournaments currently include opportunities for opponent choice:

- The EBEL (Erste Bank Eishockey Liga) Austrian Ice Hockey League (EuroHockey.com [8])
- The PRO Chess League (Klein [20])
- The (U.S.) Southern Professional Hockey League (Associated Press [1])
- The United States Bridge Federation [22]

In addition, Bleier [2] provides details of a preliminary proposal to expand the U.S. Major League Baseball (MLB) playoffs to include additional teams and permit higher-ranked teams to choose their opponents.

2.2. Related Literature

We recall that a conventional single elimination tournament design that can be represented by a symmetric binary tree with fixed seeds is conventionally termed a *bracket*. Brackets are used, for example, in regional play at the National Collegiate Athletic Association men's and women's basketball tournaments. However, the positioning of the variously ranked players allows for alternative bracket designs. This problem of bracket design, or the "seeding

problem,” has been extensively discussed in the literature. Glenn [10] compares all possible brackets with four players to evaluate the probability that each player will win, as well as the expected number of games. Searls [21] extends this analysis to eight players by comparing single and double elimination tournament designs, under single game and best-of-three-games formats. He finds that double elimination with a best-of-three-games format provides the highest probability for the best player to win and also the highest expected number of games, thereby potentially providing more revenue to the tournament designer.

Horen and Riezman [17] compare different brackets for multiple-round single elimination tournaments. They assume that the pairwise win probability matrix enables the players to be ranked such that each player ranks all the lower-ranked players in the same sequence according to the probability of beating them. We term this a *medium ranking*. They apply four fairness criteria: (i) Does the bracket maximize the probability that the best player wins the tournament? (ii) Is it order preserving—that is, no stronger player has a smaller probability of winning the tournament than a weaker player? (iii) Does it maximize the probability that the best two players meet in the final? And (iv) does it maximize the expected value of the winning player? For tournaments with four players, there are only three distinct brackets, and the conventional matchup of the strongest and weakest players at the semifinal round is the unique one that satisfies criteria (i)–(iv). For tournaments with eight players, there are 315 distinct brackets. Criterion (i) is satisfied by 8 of them. However, for criterion (ii), no bracket is satisfactory.

Vu and Shoham [25] make the stronger assumption that, for any pair of players, player P_i has a higher probability than player P_j of beating every other player. We denote this situation as the existence of a *strong ranking*. An informal description of this situation is that player P_i *dominates* player P_j . Under this assumption, they study the effect of different brackets in general size tournaments, with a view to finding a seeding that achieves order preservation (as already defined) for every such pair of players. They show that for a tournament with eight or more players, no bracket tournament design can guarantee this criterion for an arbitrary win probability matrix.

Groh et al. [12] study an elimination tournament including heterogeneous players with known abilities. Each match is studied as an auction, with equilibrium efforts characterized by mixed strategies. In any given match, win probabilities are influenced by the expected outcomes of other matches, which define expected future opponents. For tournaments with four players, it is possible to define optimal seedings to maximize the probability of the top player winning (criterion (i)) and of the final being played between the top two teams (criterion (iii)). Brown and Minor [3] consider a two-round tournament where the probability that a stronger player beats a weaker one is reduced by the previous expenditure of effort (a “spillover effect”) or by the expectation of playing a stronger future competitor at a later round (a “shadow effect”). Using data from high-stakes betting tournaments, the authors show that these two effects influence match outcomes.

Vong [23] considers the problem of strategic manipulation in multiple-round tournaments. His definition of strategic manipulation is more general than shirking. Specifically, a tournament free of strategic manipulation is characterized as one where (a) full effort exists as a subgame perfect equilibrium, and (b) each such equilibrium coincides with the outcome from a full-effort social choice function. The main result is that allowing only the top-ranked player from each group to advance to the next round of the tournament is both necessary and sufficient to achieve these properties, under an arbitrary sorting rule that sorts qualifying players from one stage to the next. However, as he notes, eliminating all but one player from every group would quickly reduce interest in the tournament and disappoint spectators who have traveled far to watch the event, as at the FIFA World Cup. This practical concern motivates our search for an alternative tournament design. If higher-ranked players are allowed to choose their opponents in the opponent choice design, then it is always better for a player to have a higher rank, *ceteris paribus*, because then it has more available choices of opponent for the next round. Although this does not fully eliminate shirking, it mitigates it in various situations.

Guyon [13] proposes the use of a *global ranking* at the elimination stage to remove group advantage, for tournaments where the number of groups is not a power of 2. Under this ranking, all teams that are qualified for the elimination stage are ranked one after another based on performance at the group stage. The proposed model was used by the Union of European Football Associations (UEFA) to modify the elimination bracket in the 2020 UEFA European Football Championship to minimize group advantage.

Guyon [14] proposes a tournament design where the players are ranked based on their earlier performance. Then, in ranked order, the surviving players choose their opponents in the next round. He also considers a variant in which players in the top half of the ranking cannot be chosen by others; that is, their high ranking affords them some protection from even higher-ranked players. Assuming that the *tournament designer* has the objective of maximizing the number of games that take place within the home country of a player, the author applies three variations of this design to data for the 2020 UEFA European Football Championship.

Our work described here differs from that of Guyon [14] in several ways. First, we consider various levels of information about pairwise win probabilities. Second, we make the natural assumption that *each player* chooses its opponents so as to *maximize its tournament win probability*. Third, we evaluate the results of the opponent choice design against the three reasonableness criteria described by Horen and Riezman [17]. Fourth, we provide anti-shirking results that improve on the conventional bracket design.

3. Proposed Design and Examples

We consider a tournament consisting of a preliminary stage, followed by a single elimination stage with multiple rounds. The single elimination stage consists of n rounds and $N = 2^n$ players. The players have a *weak ranking* at the time they enter the single elimination stage. This ranking is typically established through earlier performance, either over an entire season (e.g., the National Football League and MLB regular seasons) or at the preliminary stage (e.g., the FIFA World Cup). For a bracket tournament design—although not for the opponent choice design—this ranking completely specifies the matches that ensue for the winners at each round. The weak ranking does not imply that for any pair of players P_i and P_j , P_i has a higher probability than does P_j of beating every other player in the tournament. In practice, the strong ranking conditions specified by Vu and Shoham [25] or even the medium ranking conditions on the win probability matrix considered by Horen and Riezman [17] are not commonly found in tournaments where $N \geq 4$. We number the players in their weakly ranked order: $P_1, P_2, \dots, P_N$. Let p_{ij} denote the probability that P_i beats P_j in a single elimination play. Unless otherwise stated, we assume that all such probabilities are known to all the players. Furthermore, let $q_i^x, i = 1, \dots, N$ denote the probability that player P_i wins the tournament under a given tournament design x . If the tournament design under consideration is clear from context, we omit the superscript.

In establishing the weak ranking, there are potential issues with ties. However, all major sports have some tie-breaking rules. Where preliminary round performance provides limited information, it can be supplemented by the use of pretournament rankings. Most important, under the opponent choice design, it is always advantageous to qualify for the single elimination stage with a higher ranking.

We summarize the opponent choice design as follows.

3.1. Opponent Choice

The design we propose contains two steps.

1. Players compete in a preliminary stage, possibly lasting up to an entire season, that establishes a weak ranking among them.

Table 1. Win probability matrix for men’s professional tennis rivalries.

P_i	P_j							
	R. Nadal		R. Federer		A. Murray		J.M. Del Potro	
	Wins–losses	p_{ij}	Wins–losses	p_{ij}	Wins–losses	p_{ij}	Wins–losses	p_{ij}
N. Djokovic	30–29	0.508	27–23	0.540	25–11	0.694	16–4	0.800
R. Nadal	—	—	24–16	0.600	17–7	0.708	11–6	0.647
R. Federer			—	—	14–11	0.560	18–7	0.720
A. Murray					—	—	7–3	0.700

2. Players compete in multiple rounds of single elimination play by first choosing their opponent in weakly ranked order at each round.

As discussed in Section 1, one of the opponent choice design objectives is to motivate the players to perform well, including at the preliminary stage (i.e., to avoid shirking). The opponent choice design accomplishes this by allowing the first-ranked player to choose an opponent and similarly for other highly ranked players in ranked order.

Horen and Riezman [17], in comparing brackets with four to eight players, make several assumptions about the pairwise win probability matrix: (i) $0.5 \leq p_{ij} \leq 1$ for $1 \leq i < j \leq N$, (ii) $0 \leq p_{ij} \leq 0.5$ for $1 \leq j < i \leq N$, (iii) $p_{ij} + p_{ji} = 1$ for $1 \leq i, j \leq N$, and (iv) p_{ij} is nondecreasing in j for $1 \leq i, j \leq N$. These conditions imply that the players can be ranked such that each player ranks all the lower-ranked players in the same sequence according to the probability of beating them. If there exists a common ordering of the rows and columns of the pairwise win probability matrix such that conditions (i)–(iv) are satisfied, then the win probability matrix is said to be *strongly stochastically transitive* (David [7]), or SST.

The weak ranking we assume for the players when they enter the single elimination stage does not, in general, satisfy the SST conditions. In practice, win probability matrices are often not SST, especially for $N \geq 4$. As an example, we present data from the leading men’s professional tennis rivalries of the last 15 years, current as of July 2022 and as shown in Table 1, where the entries in row i and column j are the career wins and losses by the player in the row and the empirical estimate of p_{ij} based on those results.

In Table 1, the 4×4 submatrix involving the first four players—Novak Djokovic, Rafael Nadal, Roger Federer, and Andy Murray—is SST. However, the overall 5×5 matrix formed by the addition of Juan Martin Del Potro is not SST. This is because Djokovic has a better win record of 0.800 against Del Potro than his 0.694 record against Murray, whereas Nadal has a better win record of 0.708 against Murray than his 0.647 record against Del Potro, even though Murray is statistically a better player than Del Potro. Hence, no ordering of the players results in an SST matrix. We say that Del Potro is a “bad matchup” for Nadal, relative to their overall standards of play.

For a more extreme example of a bad matchup, consider the win probability matrix in Table 2. A conventional bracket would match P_1 and P_4 in Table 2, to the disadvantage of player P_1 . In this case, the only strategy that enables P_1 to win the tournament is to play P_2 ,

Table 2. Example of a bad matchup for player P_1 .

	P_2	P_3	P_4
P_1	1	1	0
P_2		0	0
P_3			1

Table 3. Example of an SST matrix where player P_2 dominates player P_3 .

	P_2	P_3	P_4
P_1	0.5	0.5	1
P_2		0.5	1
P_3			0.5

which enables P_3 to eliminate P_4 . Hence, player P_1 loses as a result of its high ranking, which violates the first two criteria of Horen and Riezman [17].

Remark 1. As shown by Horen and Riezman [17], given an SST win probability matrix, P_1 's bracket choice to play P_4 is optimal. It is in cases where the matrix contains non-SST irregularities that other alternatives become more attractive to P_1 .

We observe that even for an SST win probability matrix with $n = 4$, the strategy of playing P_3 may not dominate playing P_2 . Consider the SST matrix in Table 3, where the tournament win probability of player P_2 is $0.5(0.5 \times 0.5 + 0.5 \times 1) = 0.375$, whereas that of player P_3 is $0.5(1 \times 0.5 + 0 \times 1) = 0.25$.

We develop further insight from a more general situation where one difficult player should be avoided, as shown in Table 4. Clearly, player P_1 cannot choose to play against player P_4 and win the tournament. If P_1 chooses P_2 , then $q_1 = p_{12}p_{34}p_{13}$. If P_1 chooses P_3 , then $q_1 = p_{13}p_{24}p_{12}$. Therefore, P_1 should choose P_2 if and only if $p_{12}p_{34}p_{13} \geq p_{13}p_{24}p_{12}$, or equivalently $p_{34} \geq p_{24}$. This result is intuitive, because P_1 allows the other player (i.e., P_2 or P_3) that has a better chance to eliminate P_4 to do so.

Remark 2. Observe that player P_1 's choice of player P_2 or P_3 as the opponent in Table 4 is independent of whether p_{12} or p_{13} is larger. Suppose $p_{12} = p_{13} + \epsilon$, where $\epsilon > 0$ is small, but $p_{24} \gg p_{34}$. Then, player P_1 should play P_3 , even though it is easier to beat P_2 . This shows by example that player P_1 's strategy of choosing the player which it beats with the highest probability does not, in general, maximize its tournament win probability, even for $n = 4$. Thus, a myopic strategy is not optimal even in this simple case.

4. Tournament Design, Algorithm, and Data

Section 4.1 describes the opponent choice design for a single elimination tournament of general size. In Section 4.2, we describe an algorithm to determine the tournament win probability of every player, based on its ranking, under both static and dynamic designs. In Section 4.3, we describe a large data set taken from men's professional tennis tournaments. Section 4.4 provides a partial numerical example of the algorithm for a tournament with $n = 8$ players.

4.1. Design

We consider a generic single elimination stage of a tournament with n rounds and $N = 2^n$, $n = 2, 3, \dots$, players. The players have a weak ranking (i.e., a *total order*) entering the single elimination stage. At any round, player P_1 freely chooses its opponent. Then, the highest-ranked unchosen player—that is, P_2 if not chosen by P_1 , or P_3 otherwise—makes its choice, and so on. In total, in the first round, $N/2 - 1$ players choose their opponents sequentially, constituting an $(N/2 - 1)$ -stage Stackelberg game (Fudenberg and Tirole [9]). Alternative designs where fewer than $N/2 - 1$ players choose their opponents can be solved by our algorithm as special cases.

After the $N/2$ first-round games, the $N/2$ winners enter the second round. There are two alternative models for how the ranking of players and their choices are affected by earlier results. In a *static* model, a player's ranking remains the same throughout. However, in a

Table 4. Example of one difficult opponent.

	P_2	P_3	P_4
P_1	p_{12}	p_{13}	0
P_2		p_{23}	p_{24}
P_3			p_{34}

dynamic model, the winner of a game inherits the higher ranking of the two players in the game for the next round. We consider both static and dynamic rankings in our work. More generally, various ranking rules can be adopted—for example, interpolating between the two models we consider, as in the Elo system (Chess.com [5]) from chess.

We provide an example of dynamic ranking. Suppose that at some round, player P_8 beats player P_1 in the first game, and player P_5 beats player P_2 in the second game. Then, in the next round, player P_8 has a ranking of 1 and chooses its opponent first, and player P_5 has a ranking of 2 and chooses second. The rounds continue similarly, until a tournament winner is determined.

At round h , 2^{n-h+1} players play 2^{n-h} games, and $2^{n-h} - 1$ players can choose their opponents. We assume that each player has full information about the pairwise win probabilities and, if able to choose its next round opponent, does so to maximize its probability of winning the tournament.

4.2. Algorithm

We now describe our algorithm to determine each player’s choice of next-round opponent. The starting point is the final round, where each player’s opponent is fixed and its win probability is known, for every possible pair of finalists. The algorithm uses this information to determine optimal opponent choices for the semifinal round, quarterfinal round, and so on, by backward dynamic programming recursion. We describe how to solve the first-round problem. To do so, we assume that the second-round problem with $N/2$ players has been solved. That is, for any given $N/2$ players with known pairwise win probabilities, we have found the best choices for the $N/4 - 1$ players that may choose their opponents. We further assume that in the first round, a subset of players have had their opponents decided, either by their own choice or as a result of being chosen by higher-ranked players. Then, among the players with opponents not yet decided, the highest-ranked player chooses an opponent, with the objective of maximizing its tournament win probability. We describe how this player chooses its opponent. Under these assumptions, we now provide our dynamic programming algorithm for a generic n round tournament with $N = 2^n$ players.

Algorithm Opponent Choice

Input

p_{ij} , for $i, j = 1, \dots, N$, $N = 2^n$, $i \neq j$. Let $\mathcal{S}_a$ denote the set of all the players.

Value Function

For a given set $\mathcal{S}$ of players with relative rankings $1, \dots, |\mathcal{S}|$ and respective original rankings $(1), \dots, (|\mathcal{S}|)$, let $Q(\mathcal{S}) \equiv \{q_{\mathcal{S},(1)}, \dots, q_{\mathcal{S},(|\mathcal{S}|)}\}$ denote the corresponding tournament win probabilities of the players, given all players’ optimal choices of opponents.

Boundary Condition

When $|\mathcal{S}| = 2$ (i.e., only two players with relative rankings 1 and 2), we have $q_{\mathcal{S},(1)} = p_{(1)(2)}$ and $q_{\mathcal{S},(2)} = p_{(2)(1)}$, where “(1)” and “(2)” are the two players’ original rankings. Hence we assume that for any $\mathcal{S} \subseteq \mathcal{S}_a$, $|\mathcal{S}| = 2, 4, \dots, N/2$, we have found $Q(\mathcal{S}) = \{q_{\mathcal{S},(1)}, \dots, q_{\mathcal{S},(|\mathcal{S}|)}\}$.

Furthermore, when $\mathcal{S} = \mathcal{S}_a$ and the opponent of each player is decided, there are $2^{N/2}$ possible sets of the $N/2$ winners. Define ordered sets $\mathbf{X}$ and $\mathbf{X}'$ such that players in set $\mathbf{X}$ play against players in set $\mathbf{X}'$ in the first round, in the same order as in their respective sets, where $|\mathbf{X}| = |\mathbf{X}'| = N/2$. Let $\Omega(\mathbf{X}, \mathbf{X}')$ denote the collection of all possible sets of winners, and let

p_S denote the probability of the occurrence of set $S \in \Omega(\mathbf{X}, \mathbf{X}')$. Then, the tournament win probability $q_j = q_{S_a, j}$ of player P_j , $j = 1, \dots, N$, is

$$q_{S_a, j} = \sum_{S \in \Omega(\mathbf{X}, \mathbf{X}')} p_S q_{S, j}. \quad (1)$$

Optimal Solution Value $Q(S_a)$

Recurrence Relation

At any round, the algorithm works by evaluating all subsets of smaller size before all subsets of larger size. Suppose the opponents of $N - 2$ players are already decided, whereas those of two players P_i and P_j , where $i < j$, are not. Then, player P_i will play player P_j as its opponent for any $1 \leq i < j \leq N$. Next, the algorithm evaluates all subsets of four players using the previously computed information for any pair of the last two players, and so on, until the optimal choices of all N players and their win probabilities are determined. As in the boundary condition, we define ordered sets $\mathbf{X}$ and $\mathbf{X}'$ with $|\mathbf{X}| = |\mathbf{X}'|$, such that players in set $\mathbf{X}$ play against players in set $\mathbf{X}'$, in the same order as in their respective sets. By assumption, the collection of all possible sets of winners $\Omega(\mathbf{X}, \mathbf{X}')$ is known for $|\mathbf{X}| = |\mathbf{X}'| \geq m$ for some $1 \leq m \leq N/2$. Next, we assume that $2m - 2$ players have their opponents decided in ordered sets $\mathbf{X}$ and $\mathbf{X}'$ with $|\mathbf{X}| = |\mathbf{X}'| = m - 1$. Let set $\mathcal{U}$ contain all the players with undecided opponents, where player P_j has the highest ranking in $\mathcal{U}$, and $\mathbf{X} \cup \mathbf{X}' \cup \mathcal{U} = S_a$. Then, player P_j chooses its opponent to maximize its tournament win probability $q_j = q_{S_a, j}$ as follows:

$$q_{S_a, j} = \max_{i \in \mathcal{U}} \sum_{S \in \Omega(\mathbf{X} \cup \{j\}, \mathbf{X}' \cup \{i\})} p_S q_{S, j}, \quad (2)$$

where $\Omega(\mathbf{X} \cup \{j\}, \mathbf{X}' \cup \{i\})$ is known by assumption, because $|\mathbf{X} \cup \{j\}| = |\mathbf{X}' \cup \{i\}| = m$.

The main theoretical result for **Algorithm Opponent Choice** now follows.

Theorem 1. *Algorithm Opponent Choice finds the optimal sequence of opponent choices for all $N = 2^n$ players, and their tournament win probabilities, in $O((2^n)!)$ time.*

Proof. From the recurrence relation, the algorithm

- (i) considers all possible transitions from each round to the previous round, using Equation (1), and
- (ii) considers all possible transitions at each round, from the case where $2m$ players have decided opponents to the case where $2m - 2$ players have decided opponents using Equation (2), for all relevant values of m .

Therefore, the algorithm compares the tournament win probabilities of all possible states and finds an optimal solution. Regarding the time complexity, we refer the reader to Hall and Liu [16]. Q.E.D.

As an example of running time, using a laptop computer with Intel Core 2.10 GHz CPU and 8 GB RAM, an instance with 8 players can be solved within 1 second and an instance with 16 players can be solved within 12 hours. For example, the opponent choice design could be used for the single elimination stage of the FIFA World Cup, which includes 16 players.

4.3. Data

In order to test the opponent choice design in real sports tournaments, we use data from Data Hub [6] for the Association of Tennis Professionals men's tennis tournaments from 1991 to 2016. This data set includes 95,360 tennis matches over 1,902 tournaments played during that period.

To focus on matches where players seeded from no. 1 to no. 16 potentially play each other, we restrict the data set to the 28,534 matches that were played at the round-of-16 or a later round in those tournaments, where such matches are possible. These potential matches

actually occur in 5,943 instances, which are won by the higher-seeded (respectively, lower-seeded) player in 3,788 (respectively, 2,155) instances. We use the results of these 5,943 matches to estimate win probabilities between all pairs of players seeded between no. 1 and no. 16, based on win frequency. For example, there are 175 matches between the no. 1 and no. 2 seeds, with the no.1 seed winning 97 of them, thus giving an empirically estimated win probability for the no. 1 seed of $97/175 = 0.554$. In case the number of matches between two seeds is very small, we add the same number to the number of wins of each player to ensure that the number of wins by each seed is not less than 5. For example, there are 2 matches between the 15 and 16 seeds, with the no. 15 seed winning both of them; after this adjustment, we have the no. 15 seed winning 7 of the 12 matches between the two seeds. The pairwise win probability matrix is shown in Table 5.

4.4. Example of the Algorithm

We provide a numerical example of Algorithm Opponent Choice, using a dynamic ranking of players, for the quarterfinal round of a tournament that includes the top eight seeded tennis players in the data set described in Section 4.3. The algorithm starts from all the possible pairwise win probabilities for those eight players, which are used to define the boundary conditions at the final round. It then calculates the tournament win probabilities for each player, for all possible semifinal configurations. The algorithm requires that those calculations have already been completed, and we show how they are used at the quarterfinal round.

The algorithm calculates the no. 1 seed’s tournament win probability for each of its seven possible choices of quarterfinal round opponent (i.e., the 2–8 seeds). For conciseness, we show in Table 6 only the detailed calculation of the no. 1 seed’s tournament win probability if it chooses to play against the no. 7 seed, which is the choice that maximizes its tournament win probability. This probability is computed by the algorithm as follows. If the no. 1 seed makes this choice, the other quarterfinal games, given optimal sequential selections by the next highest-ranked available players which are the 2 and 3 seeds accordingly, are (2, 4), (3, 6), and (5, 8). Given a natural assumption of independence, the probability of any combination of winners from those four games is the product of their pairwise win probabilities. For example, using data from Table 5, the probability that the winners of the four quarterfinal games are the 1, 2, 3, and 5 seeds is $p_{17} \times p_{24} \times p_{36} \times p_{58} = 0.763 \times 0.690 \times 0.617 \times 0.333 = 0.108$. Computing the probabilities of all eight possible combinations of quarterfinal winners provides the results shown in the third column of Table 6. We observe that these probabilities do not sum

Table 5. Win probability matrix for seeded tennis players.

	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	0.554	0.657	0.665	0.727	0.713	0.763	0.721	0.763	0.795	0.783	0.813	0.806	0.886	0.813	0.773
2		0.669	0.690	0.621	0.630	0.603	0.670	0.784	0.650	0.650	0.844	0.718	0.833	0.817	0.773
3			0.436	0.549	0.617	0.626	0.578	0.650	0.438	0.633	0.548	0.703	0.583	0.735	0.690
4				0.503	0.506	0.667	0.636	0.533	0.750	0.412	0.538	0.733	0.593	0.607	0.815
5					0.484	0.607	0.333	0.512	0.667	0.600	0.634	0.688	0.500	0.667	0.545
6						0.519	0.560	0.448	0.567	0.596	0.643	0.500	0.560	0.643	0.722
7							0.409	0.583	0.617	0.579	0.565	0.385	0.545	0.500	0.583
8								0.483	0.532	0.643	0.379	0.385	0.455	0.583	0.706
9									0.583	0.455	0.615	0.500	0.545	0.583	0.385
10										0.545	0.706	0.357	0.545	0.545	0.583
11												0.500	0.583	0.583	0.357
12													0.615	0.667	0.357
13														0.500	0.417
14															0.500
15															0.583

Table 6. Tournament win probability of the no. 1 seed if the no. 7 seed is its quarterfinal opponent.

i	s_i	$p(s_i)$	$p(q_1 s_i)$	$p(q_1, s_i)$
1	(1, 5, 2, 3)	0.108	0.427	0.046
2	(1, 8, 2, 3)	0.217	0.424	0.092
3	(1, 5, 2, 6)	0.067	0.445	0.030
4	(1, 8, 2, 6)	0.134	0.442	0.059
5	(1, 5, 4, 3)	0.049	0.481	0.023
6	(1, 8, 4, 3)	0.097	0.477	0.046
7	(1, 5, 4, 6)	0.030	0.500	0.015
8	(1, 8, 4, 6)	0.060	0.496	0.030
			Total	0.342

to 1, because only outcomes where the no. 1 seed wins its quarterfinal game are considered. On the condition that the other semifinal round players are the no. 2, 3, and 5 seeds, the no. 1 seed has three choices:

- Play the no. 2 seed: $q_1 = 0.554[(0.549 \times 0.657) + (0.451 \times 0.727)] = 0.382$.
- Play the no. 3 seed: $q_1 = 0.657[(0.621 \times 0.554) + (0.379 \times 0.727)] = 0.407$.
- Play the no. 5 seed: $q_1 = 0.727[(0.669 \times 0.554) + (0.331 \times 0.657)] = 0.427$.

Because **Algorithm Opponent Choice** operates by backward recursion, these probabilities have already been computed by the algorithm at the semifinal round and are known to the no. 1 seed at the time of choosing its quarterfinal opponent. In view of the no. 1 seed’s objective of maximizing its tournament win probability, it chooses to play the no. 5 seed at the semifinal round.

There are eight possible semifinal configurations in which the no. 1 seed is included, as shown in the second column of Table 6. We denote these eight possible semifinals by s_i , $i = 1, \dots, 8$. The probabilities of their occurrence appear in the third column as $p(s_i)$. The conditional probability for the no. 1 seed to win the tournament from each semifinal configuration appears in the fourth column as $p(q_1 | s_i)$. The fifth column of Table 6 shows the joint probability $p(q_1, s_i) = p(s_i) \cdot p(q_1 | s_i)$ of each possible semifinal configuration i in the row and the no. 1 seed winning the tournament from that configuration. Finally, the probability of the no. 1 seed winning the tournament, still conditional on choosing the no. 7 seed as its quarterfinal opponent, appears at the bottom of the fifth column of Table 6, where $q_1 = \sum_{i=1}^8 p(q_1, s_i)$.

5. Computational Study

In this section, we use computational studies to compare the performance of the opponent choice design, under both static and dynamic rankings, to that under a conventional bracket design. Specifically, we evaluate whether the opponent choice design (a) provides a reasonably higher win probability to the top-ranked player(s) than a bracket does, (b) provides a reasonably higher probability of the top two players meeting than a bracket does, and (c) provides a ranking by tournament win probabilities that closely matches the original ranking of the players. Section 5.1 computes the tournament win probabilities using the opponent choice design for generic seeded tennis players at the semifinal and quarterfinal rounds and round-of-16, using the data set of Section 4.3, and compares them with the exact probabilities obtained from a conventional bracket design. In Section 5.2, we study the sensitivity of results from the opponent choice design to increasing irregularity in a pairwise win probability matrix on the above-mentioned performance measures (a), (b), and (c).

5.1. Win Probabilities by Seeding

Using the seeding of the tennis players as the ranking, we compute the tournament win probabilities of the various players at the semifinal round, quarterfinal round, and round-of-16, as

shown in Tables 7, 8, and 9, respectively. These probabilities are computed by assuming that the highest-ranked player(s) select the opponent(s) that maximize their tournament win probability. For example, at the semifinal round, the no. 1 seed has a tournament win probability of 0.391 if choosing to play the no. 4 seed, compared with 0.366 against the no. 2 seed and 0.386 against the no. 3 seed; it therefore chooses to play the no. 4 seed. Consequently, the other semifinal match is between the no. 2 and 3 seeds. In Table 7, the various players have identical tournament win probabilities for the bracket design and the opponent choice design.

Table 8 contains quarterfinal results, where q_i^D and q_i^S are the tournament win probabilities of player i under dynamic and static rankings of players, respectively. For the opponent choice design, each player's quarterfinal round opponent is also shown. For this example, each player's quarterfinal round opponents are the same under the dynamic and static rankings of players. Also, q_i^B is the tournament win probability of player i using the conventional quarterfinal round bracket (1, 8, 4, 5, 3, 6, 2, 7). We discuss the results in Table 8. For this instance, the no. 1 and 2 seeds both gain substantially from the opponent choice design by being able to choose their opponents. However, the no. 3 and 4 seeds have a lower tournament win probability because of the chance of being chosen by a higher-ranked player, whereas this possibility does not arise at the quarterfinal round under a conventional bracket design. The no. 5–7 seeds also incur a lower tournament win probability. The no. 8 seed gains under the opponent choice design because the no. 1 seed is not necessarily its quarterfinal opponent, as would be the case under the bracket design.

Table 9 contains round-of-16 results in a similar format. For this instance, using the opponent choice design, each player's round-of-16 opponents are the same under dynamic and static rankings of players. The main winner from the opponent choice design is the no. 1 seed, because of its first-mover advantage. The no. 3 and 4 seeds benefit from the opponent choice design with dynamic ranking, and the no. 2, 3, 4, and 5 seeds benefit from the opponent choice design with static ranking. The benefit they receive is paid for quite evenly by the no. 6–16 seeds. A particularly interesting case is the no. 14 seed, which is chosen by the no. 1 seed. This difficult matchup reduces its tournament win probability from 0.011 using the bracket design to 0.005 using the opponent choice design under static ranking but only to 0.009 under dynamic ranking. The difference in these last two numbers results from the possibility that, by beating the no. 1 seed, the no. 14 seed achieves the no. 1 ranking at the quarterfinal round under dynamic ranking, thereby providing some attractive options for the following rounds.

5.2. Sensitivity to Matrix Irregularity

We compare the effect of increasing but known irregularity in the pairwise win probability matrix on (a) the tournament win probabilities of the various players, (b) the probability that players P_1 and P_2 meet, and (c) order preservation relative to ranking. We compare the opponent choice design under both static and dynamic rankings with a conventional bracket design, for both the semifinal and quarterfinal rounds.

We start from a regular SST matrix with $N = 2^n$ players, where $p_{ij} = 0.5 + 0.05(j - i)$ for $1 \leq i < j \leq N$. We control the degree of irregularity δ by adding to each matrix entry a randomly generated value $d \sim U[-\delta, \delta]$ for $\delta = 0, 0.01, 0.02, \dots, 0.1$. We generate the value of d independently for each matrix entry. For the semifinal and quarterfinal rounds with $\delta > 0$, we have 6 and 28 random variables, respectively, and hence we use a sample size of 10,000 for

Table 7. Tournament win probabilities for seeded tennis players at the semifinal round.

	Seed i vs. opponent			
	1 vs. 4	2 vs. 3	3 vs. 2	4 vs. 1
q_i	0.391	0.353	0.124	0.132

Table 8. Tournament win probabilities for seeded tennis players at the quarterfinal round.

	Seed i vs. opponent							
	1 vs. 7	2 vs. 4	3 vs. 6	4 vs. 2	5 vs. 8	6 vs. 3	7 vs. 1	8 vs. 5
q_i^D	0.342	0.242	0.101	0.078	0.049	0.064	0.039	0.085
q_i^S	0.342	0.242	0.107	0.078	0.050	0.064	0.037	0.080
q_i^B	0.315	0.201	0.121	0.109	0.081	0.072	0.044	0.056

each δ . For the semifinal round, the dynamic and static rankings are the same under the opponent choice design because only player P_1 can choose its opponent, whereas for the quarterfinal round, they are different.

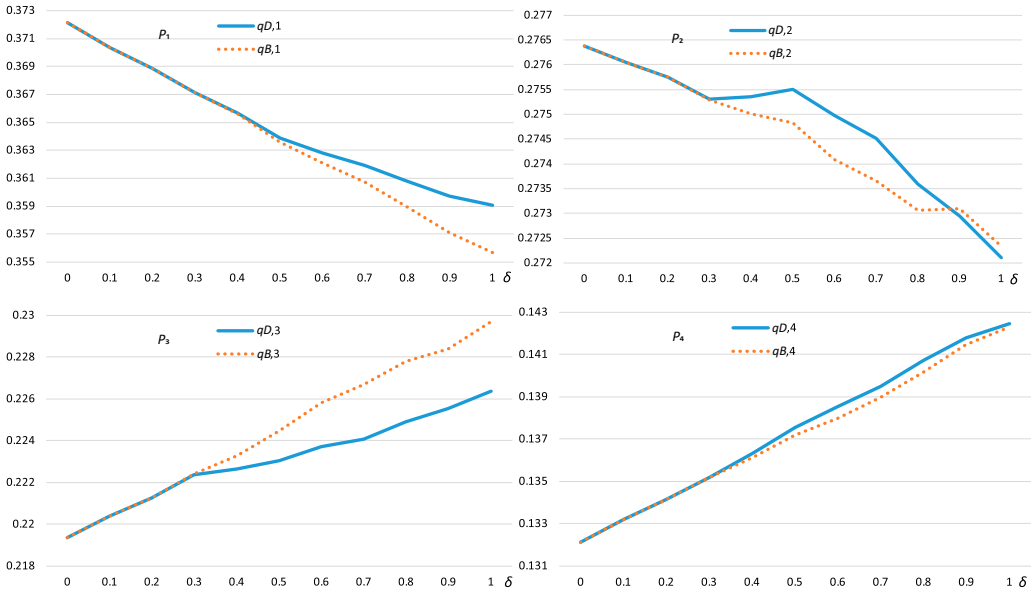
5.2.1. Tournament Win Probabilities. We study the effect of increasing irregularity on the tournament win probabilities of players with various rankings. In Figures 1 and 2, the horizontal axis represents the degree of irregularity δ , and the vertical axis represents the mean tournament win probability of a specific player. For ease of presentation, the subgraphs show different ranges on the vertical axis. In Figure 1, the solid and dotted lines represent tournament win probabilities using the opponent choice design (q_i^D, i) and using a conventional bracket design (q_i^B, i), respectively.

We discuss the semifinal round results shown in Figure 1. For the initial case, when $\delta \leq 0.025$, the win probability matrix is SST, and thus the conventional bracket design maximizes the tournament win probability of player P_1 (as shown by Horen and Riezman [17]) and performs the same as the opponent choice design for all the players. Under both designs, the tournament win probabilities of players P_1 and P_2 decrease as matrix irregularity increases, whereas those of players P_3 and P_4 increase with matrix irregularity. This is intuitive, because greater matrix irregularity gives lower-ranked players a higher probability of winning a game. The opponent choice design benefits player P_1 the most relative to the conventional bracket design when matrix irregularity is high, because the design gives player P_1 an opportunity to choose an opponent besides player P_4 . Player P_2 also benefits slightly from the opponent choice design when the matrix irregularity is moderate. This is because, with moderate irregularity, player P_1 is more likely to choose player P_3 , and consequently, player P_2 plays P_4 , which is, on average, better for player P_2 . However, when matrix irregularity is high, player P_2 may be chosen by player P_1 , and as a result, P_2 's tournament win probability is slightly less than that using the conventional bracket. Player P_3 's tournament win probability is reduced by the opponent choice design because of the nonzero probability that it is chosen by player P_1 . Finally, the opponent choice design favors player P_4 slightly, because its probability of being chosen by player P_1 is reduced.

Table 9. Tournament win probabilities for seeded tennis players at round-of-16.

	Seed i vs. opponent							
	1 vs. 14	2 vs. 12	3 vs. 15	4 vs. 16	5 vs. 13	6 vs. 11	7 vs. 10	8 vs. 9
q_i^D	0.337	0.223	0.086	0.080	0.056	0.047	0.031	0.036
q_i^S	0.336	0.232	0.089	0.085	0.057	0.047	0.030	0.035
q_i^B	0.313	0.228	0.073	0.079	0.056	0.050	0.035	0.037
	9 vs. 8	10 vs. 7	11 vs. 6	12 vs. 2	13 vs. 5	14 vs. 1	15 vs. 3	16 vs. 4
q_i^D	0.023	0.016	0.020	0.009	0.012	0.009	0.009	0.007
q_i^S	0.021	0.014	0.020	0.008	0.010	0.005	0.006	0.006
q_i^B	0.025	0.017	0.021	0.013	0.013	0.011	0.010	0.018

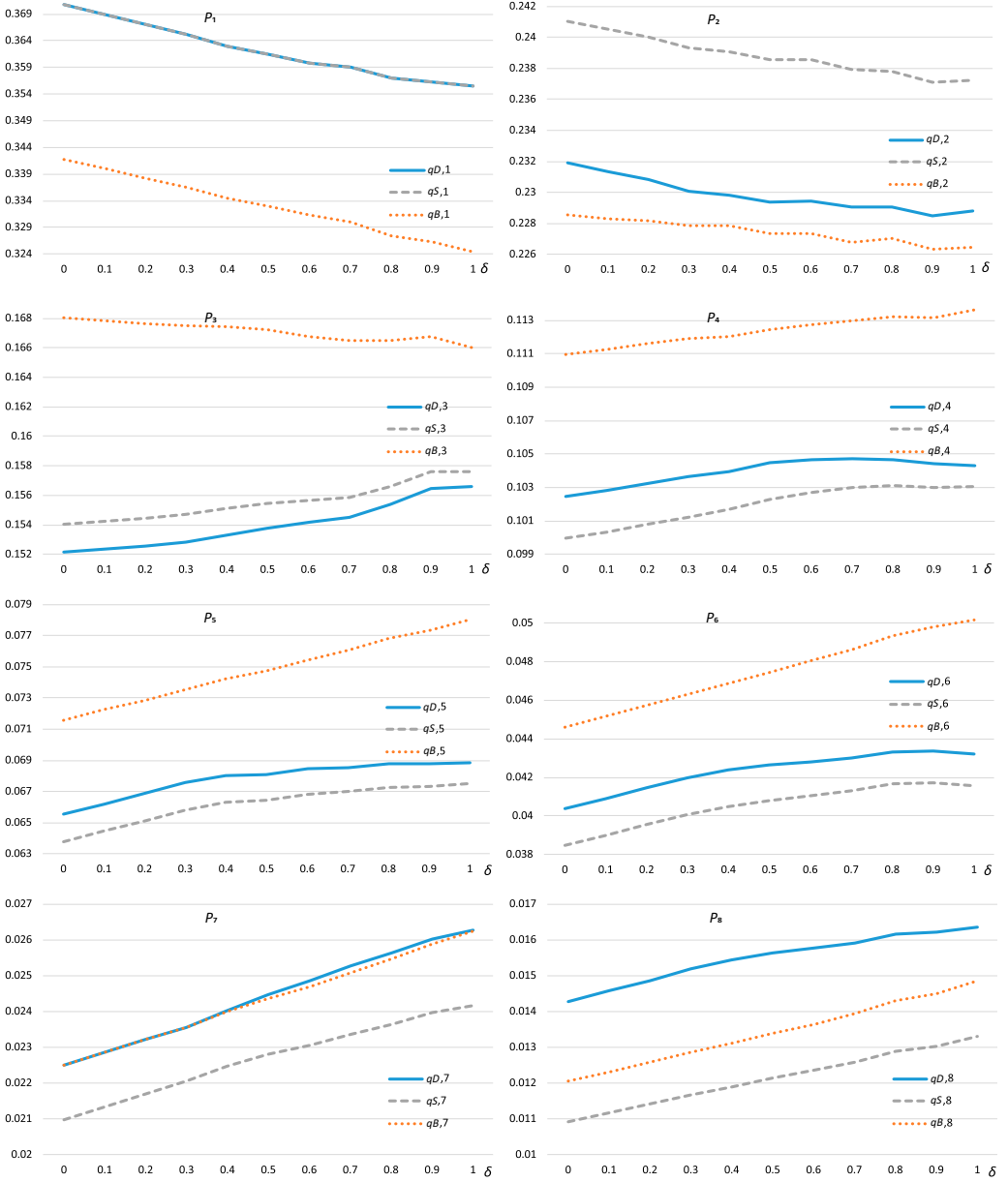
Figure 1. (Color online) Tournament win probabilities for the semifinal round.



In Figure 2, the solid and dashed lines represent tournament win probabilities using the opponent choice design under the dynamic (qD, i) and static (qS, i) designs, respectively, and the dotted line represents tournament win probabilities using a conventional bracket design (qB, i). We discuss the quarterfinal round results shown in Figure 2. The opponent choice design benefits player P_1 similarly under the dynamic and static rankings, and the two lines are indistinguishable in Figure 2. Moreover, the relative benefit is highest when matrix irregularity is high, because it is then that the value of choosing an opponent other than P_8 is greatest. If player P_1 loses and player P_2 wins different games at the quarterfinal round, then at the semifinal round under the static ranking, player P_2 chooses its opponent first, whereas under the dynamic ranking, the player that beats player P_1 in the quarterfinal round, which is likely to be a lowly ranked player, chooses its opponent first. Hence, player P_2 benefits greatly from the flexibility of choice under the static ranking but less under the dynamic ranking. The results for player P_8 are opposite to those for player P_2 . Players P_3 and P_4 's win probabilities are reduced under the opponent choice design because they may face stronger opponents as a result of the choice(s) of players P_1 and P_2 . Because player P_4 is unlikely to choose at the quarterfinal round, it benefits from the chance to improve its ranking dynamically. Finally, using the opponent choice design under dynamic ranking, each player may have an opportunity to choose its opponents. But under static ranking, players P_4 and P_5 have limited opportunities, and players P_6 – P_8 have no opportunity to choose opponents, and they are subject to the choices of higher-ranked players. Consequently, players P_4 – P_8 have consistently lower tournament win probabilities under static ranking than under dynamic ranking.

5.2.2. Top Two Ranked Players Meeting. As noted by Horen and Riezman [17], a desirable feature of tournament design is the probability that it provides for the top two ranked players to meet; such matchups typically generate great spectator interest and strong revenue for the tournament designer. We compare the probabilities for players P_1 and P_2 to meet under different designs. Under the conventional bracket, players P_1 and P_2 can only

Figure 2. (Color online) Tournament win probabilities for the quarterfinal round.



meet in the final, whereas under the opponent choice design, they can meet in any round. Our instances contain no cases where players P_1 and P_2 meet in the quarterfinal round; hence we report the probabilities for them to meet in the semifinal and final rounds.

In Table 10, columns $q_{D4,12}$, $q'_{D4,12}$, and $q_{B4,12}$ contain the mean probabilities that players P_1 and P_2 meet in the final under the opponent choice design, in the semifinal round under the opponent choice design, and in the final round under the conventional bracket design, respectively, for the semifinal round. Similarly, the next five columns apply to the quarterfinal round, with the dynamic ($q_{D8,12}$, $q'_{D8,12}$) and static ($q_{S8,12}$, $q'_{S8,12}$) rankings separated. We observe that with small matrix irregularity $\delta \leq 0.03$, the designs perform similarly. But with

Table 10. Probability of player P_1 meeting player P_2 .

Parameter	$q_{D4,12}$	$q'_{D4,12}$	$q_{B4,12}$	$q_{D8,12}$	$q'_{D8,12}$	$q_{S8,12}$	$q'_{S8,12}$	$q_{B8,12}$
$\delta = 0$	0.358	0.000	0.358	0.257	0.000	0.257	0.000	0.258
$\delta = 0.01$	0.356	0.000	0.356	0.256	0.000	0.256	0.000	0.256
$\delta = 0.02$	0.354	0.000	0.354	0.254	0.000	0.254	0.000	0.254
$\delta = 0.03$	0.353	0.000	0.352	0.252	0.000	0.252	0.000	0.252
$\delta = 0.04$	0.352	0.000	0.351	0.250	0.000	0.250	0.000	0.251
$\delta = 0.05$	0.352	0.000	0.349	0.250	0.000	0.250	0.000	0.249
$\delta = 0.06$	0.351	0.002	0.348	0.249	0.001	0.249	0.001	0.247
$\delta = 0.07$	0.348	0.013	0.346	0.249	0.003	0.249	0.003	0.245
$\delta = 0.08$	0.345	0.022	0.344	0.248	0.007	0.248	0.007	0.243
$\delta = 0.09$	0.339	0.039	0.343	0.247	0.011	0.247	0.011	0.241
$\delta = 0.10$	0.337	0.049	0.341	0.247	0.015	0.247	0.015	0.240
Mean	0.349	0.011	0.349	0.251	0.003	0.251	0.003	0.249

increased irregularity, the opponent choice design achieves a greater probability for players P_1 and P_2 to meet. In the semifinal round, when the irregularity is very high ($\delta \geq 0.09$), the chance for these players to meet in the final under the opponent choice design is slightly less but is more than compensated by the probability that they meet in the semifinal round. However, in the quarterfinal round, even with high irregularity and increased probability for players P_1 and P_2 to meet in the semifinal round, the opponent choice design still offers a greater probability for them to meet in the final than the conventional bracket design does.

5.2.3. Rank Preservation. We examine how the ranking of players by their tournament win probability under each design preserves their original ranking. Recall that preservation of ranking is a desirable property of tournament design (Horen and Riezman [17], Vu and Shoham [25]). For this performance measure, the bracket design performs well, but we show that the opponent choice design produces very similar results. For each design and δ value, we compute the *Kendall tau rank correlation coefficient* (Kendall [19]), which we now define. Let $(x_1, y_1), \dots, (x_N, y_N)$ be a set of different rankings. A pair of rankings (x_i, y_i) and (x_j, y_j) are *concordant* if $x_i > x_j$ and $y_i > y_j$, or $x_i < x_j$ and $y_i < y_j$; they are *discordant* if $x_i > x_j$ and $y_i < y_j$, or $x_i < x_j$ and $y_i > y_j$. Specifically, the Kendall coefficient τ , where $-1 \leq \tau \leq 1$, is defined as

$$\tau = \frac{2(\text{number of concordant pairs} - \text{number of discordant pairs})}{n(n-1)}.$$

The results are summarized in Table 11. For the semifinal round, columns τ_{D4} and τ_{B4} represent the Kendall coefficients between the players' original ranking and their ranking by tournament win probability, as found by the opponent choice design and the conventional bracket, respectively. For the quarterfinal round, columns τ_{D8} , τ_{S8} , and τ_{B8} represent the Kendall coefficients between the players' original ranking and their ranking by tournament win probability found by the opponent choice design with dynamic ranking, the opponent choice design with static ranking, and the conventional bracket, respectively. The rows correspond to different values of the irregularity parameter δ . When $\delta \leq 0.03$, all the coefficients are 1.000. This confirms that for these randomly generated win probability matrices, the opponent choice design is perfectly order preserving. Table 11 starts with $\delta = 0.04$. The opponent choice design produces tournament win probabilities with similar rankings by dynamic and static ranking. With increasing irregularity δ , the opponent choice design achieves tournament win probabilities that are slightly less consistent with the players' rankings than those found by the conventional bracket. This is because the conventional bracket determines opponents strictly by ranking; hence every position in the ranking affects a player's opponent in the tournament. By contrast, in the opponent choice design, only player P_1 chooses its

Table 11. Kendall coefficient for player rankings.

Parameter	τ_{D4}	τ_{B4}	τ_{D8}	τ_{S8}	τ_{B8}
$\delta = 0.04$	0.998	0.998	1.000	1.000	1.000
$\delta = 0.05$	0.990	0.989	0.999	0.999	1.000
$\delta = 0.06$	0.970	0.970	0.996	0.997	0.999
$\delta = 0.07$	0.945	0.947	0.991	0.991	0.996
$\delta = 0.08$	0.921	0.924	0.983	0.983	0.990
$\delta = 0.09$	0.894	0.900	0.974	0.974	0.983
$\delta = 0.10$	0.870	0.876	0.964	0.964	0.973
Mean	0.941	0.943	0.987	0.987	0.991

opponent in the semifinal round, and only three players with higher seeding choose their opponents in the quarterfinal round. It should be noted, however, that with increasing irregularity, the players' original ranking becomes a less accurate measure of their relative strength. Hence, order preservation with ranking becomes less equivalent to order preservation with player strength and a less relevant performance measure.

6. Incomplete Information

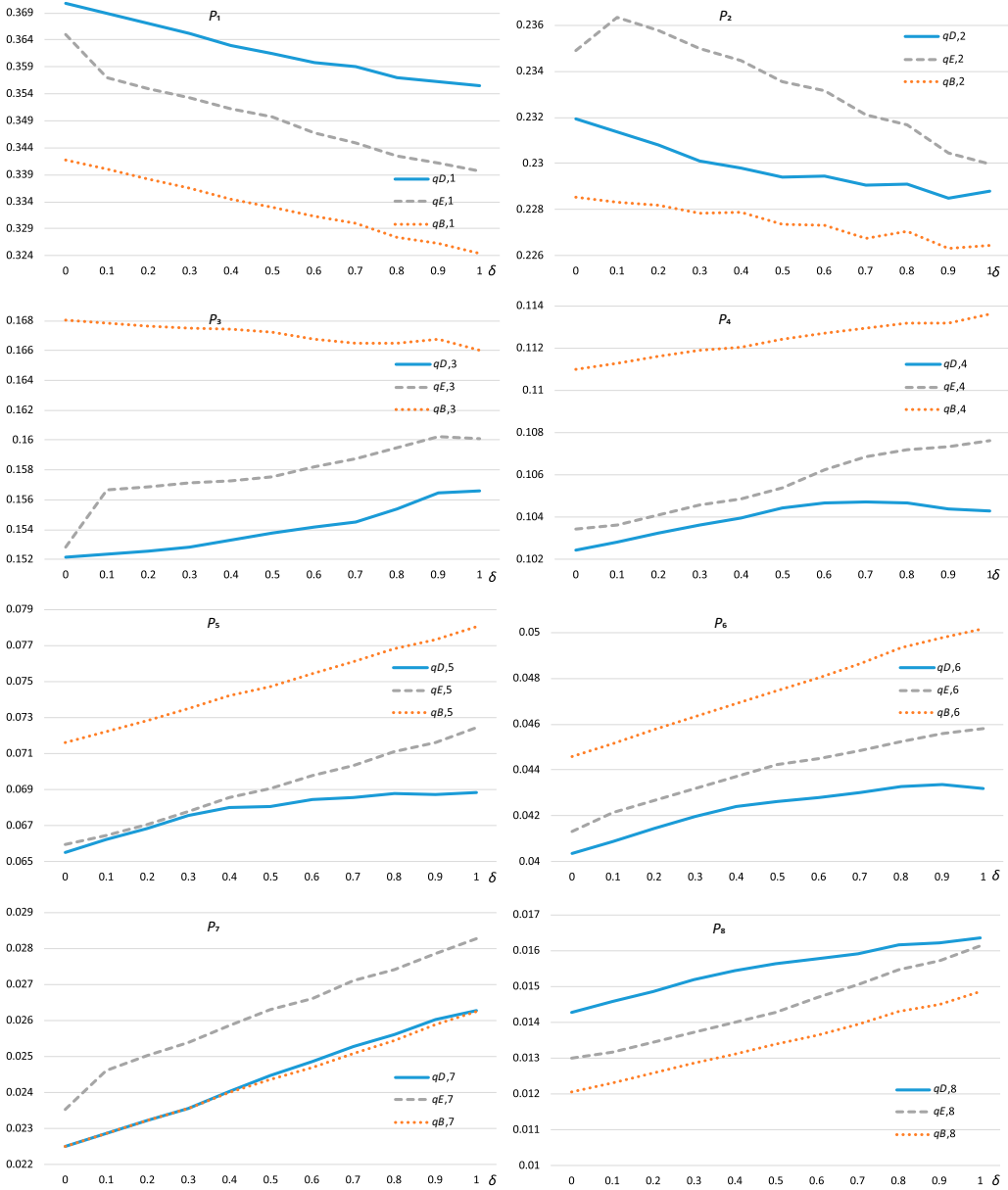
In this section, we drop the aforementioned assumption of complete information. Section 6.1 considers a case where players have imprecise probability information about pairwise win probabilities. In Section 6.2, they have no probability information at all about games between other players.

6.1. Imprecise Information

With probability information that is known to be estimated or imprecise, the players may simplify their estimates into a few categories. Consider the following model for probability estimates. If the win probability is estimated as greater than 0.6, the player uses a value of 1; if the win probability is between 0.4 and 0.6, the player uses a value of 0.5; and if the win probability is smaller than 0.4, the player uses a value of 0. We study the tournament win probabilities of the various players under incomplete information using the opponent choice design under dynamic ranking at the quarterfinal round. The exact but unknown win probability matrix is generated as in Section 5.2.1. Because of the limited number of pairwise win probabilities, ties for tournament win probability may occur in opponent choice. In such cases, we assume that each player breaks ties by choosing an opponent with the lowest ranking. Following the opponent choice of players using imprecise information, we compute the true tournament win probabilities of players using the matrix information, and we compare the probabilities with those obtained by the opponent choice design under both dynamic ranking and the conventional bracket design with complete information.

The results are shown in Figure 3. As in Figure 2, the solid line represents tournament win probabilities using the opponent choice design under dynamic ranking (qD, i), and the dotted line represents tournament win probabilities using the conventional bracket design (qB, i), both with complete information. The dashed line represents tournament win probabilities for opponent choice using the opponent choice design under dynamic ranking with imprecise information (qE, i). The results show that for all players except for P_2 and P_7 , the tournament win probabilities under opponent choice with incomplete information fall between those obtained by the other two approaches with complete information. This is intuitive: with simplified estimates of 0, 0.5, and 1 for the pairwise win probabilities, players cannot accurately differentiate the other players in their opponent choice and hence make decisions that are more similar to those under the conventional bracket design. However, players P_2 and P_7

Figure 3. (Color online) Tournament win probabilities with incomplete information.



benefit from opponent choice compared with the conventional design, even more with incomplete information than with complete information. This is because with imprecise information, the first-mover player P_1 cannot make opponent choice decisions that are as accurate as in the case of complete information. Hence, the second-mover player P_2 improves its tournament win probability. Player P_7 benefits from opponent choice with imprecise information because it is less likely to be chosen by player P_1 than under opponent choice with full information and by player P_2 than under the conventional bracket design. We also note that player P_2 's tournament win probability is not monotonic in the level of matrix irregularity δ under imprecise information. This is because when $\delta = 0$, the estimated pairwise win probabilities are close to their true values.

6.2. No Information

A situation with even less information than that studied in Section 6.1 arises when player P_1 has knowledge only of its own win probabilities and no probability information about matches between pairs of the other players. For this case, we have the following result.

Theorem 2. *For any instance of an n -stage tournament, let players $P_2, \dots, P_{2^n}$ be ordered such that $p_{12} \leq p_{13} \leq \dots \leq p_{12^n}$. Then, if player P_1 chooses the lowest-ranked available player at each round, this strategy achieves $q_1 \geq \prod_{i=1}^n p_{12^i}$, and this bound is tight.*

Proof. We prove the result by induction on the number of rounds played. Let $q_1(k)$ denote the probability that player P_1 wins all its matches at the first k rounds. For the basis where $k = 1$, player P_1 chooses to play against player P_{2^n} and wins with probability p_{12^n} . For the induction hypothesis, we assume that the result holds for rounds $1, \dots, k$ and prove that it holds for round $k + 1$. Thus, we assume that $q_1(k) \geq \prod_{i=n-k+1}^n p_{12^i}$. The induction step is completed by observing that, at round $k + 1$, there must exist a remaining player with an index of at least 2^{n-k} . If player P_1 chooses to play against this player, then we have $q_1(k+1) \geq \prod_{i=n-k+1}^n p_{12^i} \cdot p_{12^{n-k}} = \prod_{i=n-k}^n p_{12^i}$.

To show that the bound is tight, recall that, by assumption, player P_1 does not know the win probabilities of other pairs of players. Let $p_{ij} = 1$ for $2 \leq i < j \leq 2^n$. In this case, when k rounds remain to be played, the remaining players are exactly $P_1, \dots, P_{2^k}$. Therefore, there is no possibility for player P_1 to achieve a higher value of q_1 . Q.E.D.

However, we observe that Theorem 2 does not imply that player P_1 , in general, maximizes q_1 by choosing to play the player against which P_1 has the highest win probability. Indeed, Remark 2 in Section 3 provides a counterexample to such myopic strategies, even for $n = 4$.

7. Shirking

When players shirk, there are serious implications for the credibility and prestige of the tournament—and, in some situations, also for national pride. In addition, major sports are a venue for substantial betting, which raises the possibility of legal liability for fraud when teams shirk.

In this section, we study the issue of shirking under both individual and group strategies by the players. Section 7.1 reviews several controversial incidents of shirking. Section 7.2 discusses the extent to which our opponent choice design can reduce shirking against individual strategic behavior. Section 7.3 contains a similar discussion for group strategic behavior.

7.1. Examples of Shirking

We list some famous examples of shirking in sports tournaments. See Vampiew [23] and the references cited in the following items:

- In the 1998 Tiger Cup Asian football competition, the winner of the last group game between Thailand and Indonesia would travel to Hanoi to play the host Vietnam, whereas the loser could stay in Ho Chi Minh City and play Singapore. Some apathetic play by both teams ended when an Indonesian player egregiously scored for Thailand in extra time, which resulted in a lifetime ban from international football.
- The NBA changed its postseason tournament design after the 2005–2006 Los Angeles Clippers apparently lost some late regular season games deliberately. Their incentive to do so was to avoid facing the Dallas Mavericks in the first round of the playoffs (Graham [11]).
- In the ice hockey competition at the 2006 Winter Olympics, the Swedish coach publicly discussed the benefit of losing an upcoming match with Slovakia, in order to avoid playing either the Czech Republic or Canada, the two most recent gold medalists, in the next round. Sweden lost the match 3–0 and went on to win the gold medal (Burnside [4]).

- At the 2012 Summer Olympics, the Chinese, Indonesian, and South Korean women's badminton teams were all disqualified for deliberately losing their group stage matches in order to face easier later matches (Hurst [18]).
- At the 2018 FIFA World Cup, the last group match involved England and Belgium. The winner would face a potential quarterfinal match against five-time world champions Brazil. Both teams rested their top players. Belgium won the match 1–0 but later beat Brazil anyway (Haisley [15]).

7.2. Individual Strategic Behavior

We discuss the extent to which our opponent choice tournament design reduces the players' incentive to shirk under strategic behavior by individual players.

Remark 3. From Vong [24], it is not possible to eliminate shirking entirely without imposing on the tournament design a severe restriction that may reduce the tournament's appeal to spectators.

From Remark 3, we specifically consider the final game at the end of the preliminary round. Although this restricts the scope of our discussion, it is in exactly this situation that shirking is most likely. Moreover, this situation includes, for example, the England versus Belgium and Thailand versus Indonesia games described earlier. We consider the incentive for a player, P , to shirk in the final game against an opponent, O . We have the following negative result.

Theorem 3. *If the subset of players that continue to the following round is dependent on the result of the final preliminary round game, then it is not possible to eliminate shirking.*

Proof. We prove by example. Suppose that by winning the final game, P eliminates O but allows into the next round another player, E , that P cannot beat. Then, there exists a win probability matrix where P 's tournament win probability is reduced by the presence of E in the next round. An example of such a matrix occurs where player E beats every player that will continue to the next round, including player P . Q.E.D.

From Theorem 3, we assume that the subset of players that continue to the next round is not dependent on the results of the last game. This situation again includes the England versus Belgium and Thailand versus Indonesia games. We provide the following anti-shirking result.

Theorem 4. *Assume that*

- the subset of players that continue to the next round is not dependent on the result of the final preliminary round game, and*
- any player in the last preliminary round game that continues to the next round will choose an opponent, rather than be chosen, at that round.*

Then, the opponent choice tournament design eliminates shirking at that game.

Proof. There are three possible cases.

(a) Player P will continue to the next round, but player O will not. A higher ranking for player P in the next round gives player P a superset of choices against a fixed set of potential opponents with a fixed ranking—and therefore a fixed order of opponent choice—relative to a lower ranking for player P . Therefore, player P has no incentive to shirk. Clearly, player O has no incentive to shirk, as this is the player's last game.

(b) Both players will continue to the next round. Both players have an incentive to improve their rankings, and therefore enlarge their choice of opponents at the next round, by trying to win the game.

(c) Neither player will continue to the next round, hence neither player has an incentive to shirk. Q.E.D.

We observe that, in the situations described in Theorem 4, our opponent choice tournament design represents an improvement over the conventional bracket design where winning the final preliminary round game may result in player P having a bad matchup in the next round, which provides an incentive to shirk at the last preliminary round game.

7.3. Group Strategic Behavior

We consider possible strategic behavior by groups of players. This is most likely to occur where a tournament contains multiple teams representing the same country—for example, in tennis, table tennis, or badminton at the Olympics. Clearly, any result that fails to eliminate shirking under individual strategic behavior also fails under group strategic behavior. Therefore, Remark 3 and Theorem 3 continue to hold in the group strategic behavior situation.

Next, we study whether the positive anti-shirking results in Theorem 4 continue to hold under group strategic behavior.

Remark 4. We note the following:

- (a) The result in part (a) of the proof of Theorem 4 partly fails under group strategic behavior. It remains true that player P will not shirk, but player O may shirk in order to provide player P with a higher ranking at the next round.
- (b) The result in part (b) of the proof of Theorem 4 also partly fails under group strategic behavior. In a situation where one player's ranking will not change depending on the results of the game, but the other player's ranking will change, the former player may shirk.
- (c) The result in part (c) of the proof of Theorem 4 still holds.

We also provide a further negative result for the group strategy case. It applies to tournaments such as soccer, where a draw is possible at preliminary round games.

Theorem 5. *Consider the last game of the preliminary round in a tournament where a draw is possible. Then, there is no design under group strategic behavior that prevents both players from shirking.*

Proof. We prove by example. Consider a win probability matrix where only a draw between players P and O will allow both players to advance, which is the only way to eliminate another player, E , which always beats all continuing players, including both P and O . Q.E.D.

8. Summary

This work addresses several deficiencies that arise in multiple-round sports tournaments. These deficiencies include poor performance with respect to reasonable criteria, the existence of an unnecessary element of chance in how matchups evolve, incentives for shirking, and the inability to integrate newly developing information. To resolve these deficiencies, we propose a new design for the single elimination stage of a tournament, where highly ranked players may choose their opponents at each round. This design is implemented for both static and dynamic rankings of the players. We describe an algorithm that computes, for each player, the optimal sequence of players to choose from those available in order to maximize its tournament win probability. Using a data set from 1,902 men's professional tennis tournaments, we demonstrate the reasonableness of the results obtained from the opponent choice design. We also study the sensitivity of the performance of the opponent choice design to increased randomness in win probabilities. An additional advantage that arises from the use of the opponent choice design is increased spectator interest from the unpredictability of matchups at future rounds, as well as potentially from a broadcast event at which opponent selections are made. Most important, an easier path through the tournament is available to any player

as a result of obtaining a higher ranking, as verified by the results from our computational study in Section 5. This feature of our opponent choice design discourages, and in some cases fully eliminates, strategic shirking behavior at the preliminary stage of the tournament.

Several interesting topics remain open for future research, as in the following list. The first topic applies our work directly, whereas the other six extend it.

- The application of the opponent choice design to empirically based studies of various sports and competitions can be explored. Examples where substantial data for pairwise matchups are available include table tennis and chess.
- In situations where the pairwise win probabilities are hard to estimate, especially when there are many players involved, a player may resort to a myopic strategy: choose an available opponent that the player can beat with the highest probability at each round. The effectiveness of myopic strategies should be studied.
- Guyon [14] proposes prohibiting the choice of other highly ranked players.
- The opponent choice design can also be used where a player's objective is to reach a particular round rather than win the entire tournament. This scenario arises toward the end of a season, where a player that is seeking to be ranked first at the end of the season needs only to reach a particular round. The sequence of choices of opponents that maximizes the probability of reaching a particular round is not, in general, the same as that which maximizes the player's tournament win probability (see Remark 2).
- It would be valuable to study how the results of a particular round could be used to modify the win probability matrix and, consequently, the choices of the players at later rounds. For example, a player that has won previous matches but with unexpectedly poor performances may become a more attractive opponent at a later round. Injuries incurred during the tournament may also affect future win probabilities.
- A natural extension of the opponent choice design under dynamic ranking would be to allow the ranking of the players to be adjusted dynamically, based on detailed performance within the tournament as measured, for example, by margin of victory. We observe that the effect of doing this may be different from the effect of adjusting the win probabilities (see the previous point).
- Some spectators of sports events enjoy the situation where one player is clearly favored over another, the latter often being termed an "underdog." It is intuitive that the use of our opponent choice design tends to reduce the *apparent* number of such matchups because there are fewer matches with extreme differences in rankings. However, because higher-ranked players are choosing their opponents based on their actual win probabilities, the *actual* number of such matchups tends to increase. This is a trade-off that should be evaluated by tournament designers, who are in the best position to understand the interests of their spectators.

In conclusion, we believe that the use of opponent choice can enhance the enjoyment of billions of sports fans around the world.

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